

Binding of Per- and Polyfluoroalkyl Substances (PFAS) to Serum Proteins: Implications for Toxicokinetics in Humans

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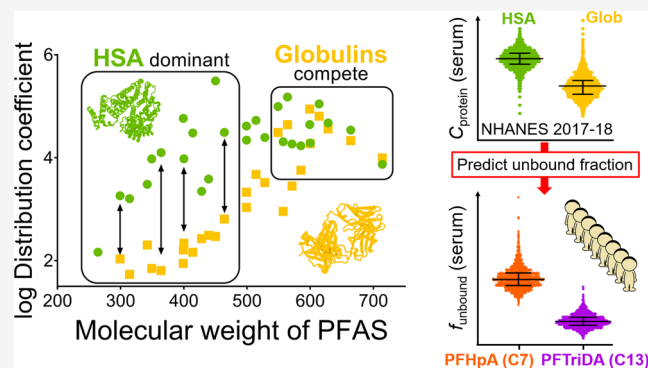
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ABSTRACT: Per- and polyfluoroalkyl substances (PFAS) are a diverse class of highly persistent anthropogenic chemicals that are detectable in the serum of most humans. PFAS exposure has been associated with many adverse effects on human health including immunotoxicity, increased risk of certain cancers, and metabolic disruption. PFAS binding to the most abundant blood serum proteins (human serum albumin [HSA] and globulins) is thought to affect transport to active sites, toxicity, and elimination half-lives. However, few studies have investigated the competitive binding of PFAS to these proteins in human serum. Here, we use C18 solid-phase microextraction fibers to measure HSA–water and globulin–water distribution coefficients ($D_{\text{HSA/w}}$, $D_{\text{glob/w}}$) for PFAS with carbon chains containing 4 to 13 perfluorinated carbons ($\eta_{\text{pfc}} = 4–13$) and several functional head-groups. PFAS with $\eta_{\text{pfc}} < 7$ were highly bound to HSA relative to globulins, whereas PFAS with $\eta_{\text{pfc}} \geq 7$ showed a greater propensity for binding to globulins. Experimentally measured $D_{\text{HSA/w}}$ and $D_{\text{glob/w}}$ and concentrations of serum proteins successfully predicted the variability in PFAS binding in human serum. We estimated that the unbound fraction of serum PFAS varied by up to a factor of 2.5 among individuals participating in the 2017–2018 U.S. National Health and Nutrition Examination Survey. These results suggest that serum HSA and globulins are important covariates for epidemiological studies aimed at understanding the effects of PFAS exposure.

KEYWORDS: exposure, unbound fraction, interindividual variability, solid-phase microextraction, human serum albumin, globulins



1. INTRODUCTION

Exposures to per- and polyfluoroalkyl substances (PFAS), a diverse class of highly fluorinated anthropogenic chemicals, have been linked to many adverse health effects in humans.^{1–6} The extreme resistance of the perfluorinated carbon bond to degradation and high production volumes used by industry and in consumer products have resulted in their ubiquitous presence in the environment and detection in the serum of 98–99% of United States (U.S.) individuals.^{7–9} PFAS binding to serum proteins is known to affect their accumulation, circulatory distribution, and elimination in humans.^{10–15} Specifically, the unbound fraction of a toxicant in blood is typically more available for transport to organs and more likely to reach molecular targets that elicit toxic effects.^{15,16} PFAS bound to proteins are thought to be inaccessible to biomolecules involved in transformation and elimination processes and are recirculated in the bloodstream, thereby extending elimination half-lives.^{17,18} Two of the most abundant proteins in blood are human serum albumin (HSA) and globulins. However, competitive binding of diverse PFAS to these proteins in human serum and implications for

toxicokinetics in the human body are poorly understood and are the focus of this work.

PFAS binding to serum proteins can be expressed by the distribution coefficient (D), which encompasses the binding of both charged and neutral PFAS to the protein.¹⁹ Past work has reported distribution coefficients for human serum albumin (HSA, i.e., $D_{\text{HSA/w}}$).²⁰ However, binding of PFAS to globulins, the second most abundant class of blood proteins, is poorly characterized (i.e., globulin–water distribution coefficients, $D_{\text{glob/w}}$), as is competitive interactions between these two proteins and PFAS. PFAS are strongly bound to HSA^{10,12–15,21,22} and $D_{\text{HSA/w}}$ generally increases with greater perfluorinated carbon-chain length (η_{pfc}) and hydrophobicity.^{11,14,23,24} These properties may explain the long elimination half-lives for longer chain PFAS ($\eta_{\text{pfc}} \geq 6$) in humans.^{25–27}

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The unbound fraction of PFAS in blood (f_{unbound}) can be estimated from the distribution coefficients for HSA and globulins and measured serum protein levels. Protein concentrations measured in human serum (i.e., the liquid component of blood after the blood has clotted) are known to vary based on environmental and longitudinal factors²⁸ and disease status.^{29,30} Quantifying the unbound fraction of PFAS in blood is useful for understanding their tissue distribution in the human body³¹ and variability in elimination half-lives for PFAS across individuals.²⁷ Epidemiological studies investigating associations between PFAS serum concentrations and health outcomes in humans may also benefit from accounting for variability in blood protein concentrations and binding affinities for different PFAS as additional covariates.

Distribution coefficients of chemicals to blood proteins and other biomolecules can be measured using solid-phase microextraction (SPME) techniques.³² The difference in chemical mass sampled by an SPME fiber in the presence and absence of biomolecules in solution represents the chemical fraction bound to the biomolecules of interest. A distribution coefficient can be derived from this fraction and the mass of the biomolecule in solution. This approach avoids extractions of chemicals from biomolecules, thereby minimizing laboratory work and matrix effects during chemical analysis. The utility of C18 SPME fibers for measuring PFAS serum protein binding has been demonstrated in past work that used them to measure binding of four PFAS to bovine serum albumin (BSA), human plasma, and *in vitro* cell homogenates.¹⁶

The main objectives of this work were to (1) investigate competitive binding of PFAS to two major proteins (HSA and γ -globulin) in human serum using SPME techniques and (2) characterize variability in the unbound fraction of PFAS in the human body. We focused on distribution of PFAS to γ -globulin because it is the most abundant protein within the class of globulins in human serum.³³ We also compared binding of PFAS to HSA and bovine serum albumin (BSA) because it is often used as a proxy for HSA.^{12,16,21} To do this, we adapted and modified a previously developed C18 SPME fiber method¹⁷ to measure distribution coefficients of PFAS with $\eta_{\text{pic}} = 4\text{--}13$ and different functional head groups (carboxylic acids, sulfonic acids, and sulfonamides) to proteins in human serum. We chose these PFAS because preliminary studies of extractable organofluorine (EOF) as a proxy for total PFAS suggest that they account for most PFAS in human serum, despite production of diverse chemicals in this class.^{34,35} We evaluated whether PFAS distribution to HSA and globulins is a reasonable predictor for the unbound fraction of PFAS in serum (f_{unbound}) by comparing experimental results to measured serum–water distribution coefficients ($D_{\text{serum/w}}$). To better understand potential variability in the unbound fraction of PFAS in blood at the population level, we examined variability in HSA and globulin in the 2017–2018 National Health and Nutrition Examination Survey (NHANES) data.⁹ We discuss how experimental distribution coefficients for blood proteins and associated proxies for the unbound fraction of PFAS in blood can be leveraged to improve toxicokinetic (TK) models and to assess susceptibility to PFAS exposures.

2. MATERIALS & METHODS

2.1. Materials and Chemicals. C18 SPME fibers with a calculated coating volume of 520 nL were purchased from

Supelco (#57281-U).³² PFAS purchased from Wellington Laboratories were tested as single chemicals or in batch experiments as mixtures. A mixture of five PFAS was used for method development, while a 32 PFAS mixture was used for protein binding experiments. For batch experiments with 32 PFAS, the PFAC–24PAR mixture from Wellington Laboratories was combined with additional PFAS that were not included in the mixture. Mass-labeled mixtures of PFAS were used as internal standards (Wellington Laboratories). Additional information on native and labeled PFAS standards is provided in the [Supporting Information \(SI, Table S1\)](#). Methanol (MeOH) and deionized water (DI) were used as solvents. Phosphate buffered saline (PBS: 0.01 M, NaCl: 0.138 M; KCl: 0.0027 M) yielding a pH of 7.4 at 25 °C was purchased from Sigma (#P3813).

Human serum albumin (Sigma, no. A1887) and γ -globulins (Sigma, no. G4386) were used as reference serum proteins. A serum standard from a human male residing in the U.S. was used for experiments and contained a total protein concentration of 40–90 g/L (Sigma, #H4522, see [Supporting Information \(SI\) Section S1](#) for product specifications).

2.2. Preparation and Storage of C18 Fibers. SPME fibers were stored in 5 mL polypropylene (PP) tubes containing 4 mL of a MeOH/DI mixture (50:50) until use in experiments. During storage, preconditioning, and experiments, the fiber holders were stabilized in prepunctured caps to avoid damaging the fiber coating (see [Section S2, Figures S1 and S2](#)). SPME fibers were incubated in 100% MeOH and 100% DI at room temperature for ≥ 2 h and ≥ 10 min, respectively, to precondition them for binding experiments before transfer to 50 mL falcon tubes for binding experiments. Rapid transfer of SPME fibers among storage, preconditioning, and experimental tubes avoided drying of the C18 polymer coating.

2.3. Setup of Binding Experiments. The molar ratio of PFAS to proteins was kept below 0.004 (HSA, BSA) and 0.009 (γ -globulin) in all binding experiments (i.e., 261 and 111 mol of protein per mol of PFAS, respectively), which is well below where saturation of serum proteins was observed in previous studies that measured binding isotherms for PFAS^{11,16,36–38} and other chemicals.^{19,24,32} For example, nonlinearity of BSA binding has been observed for PFBA, PFOA, PFHxS, and PFOS at molar ratios of ≥ 0.39 ,¹⁶ emphasizing that our experiments were conducted at concentrations within the linear binding range. Low molar ratios were selected to mimic binding in serum at PFAS concentrations relevant to human exposure, based on ratios of ≤ 0.0005 that were estimated from PFAS concentrations reported in human sera.⁹

The BSA, HSA, γ -globulins, and human serum were dissolved in defined amounts (varied among experiments) in 40 mL of PBS in a 50 mL falcon tube and incubated in a horizontal shaker at 37 °C and 100 rpm for ≥ 15 min. A 40 mL PBS solution free of proteins/serum was prepared and treated similarly as a control. Defined amounts of PFAS stock solutions in MeOH were added to 50 mL falcon tubes and evaporated to dryness to avoid MeOH contamination in PBS samples. The PBS and protein solutions were added to the falcon tubes and incubated on the horizontal shaker at 37 °C and 100 rpm for ≥ 1 h. The solution was evenly distributed in 4 mL aliquots to PP tubes ($n = 3$ to 5 replicates). Preconditioned SPME fibers were transferred to the tubes while ensuring that the fiber coating was completely immersed in the liquid. Recovery controls containing PFAS in PBS without SPME

fibers were prepared to assess loss processes other than binding to the SPME fiber (e.g., binding to vial materials, degradation). Fiber controls containing PBS, proteins, and serum without PFAS were prepared to assess PFAS background and cross contamination between vials and/or experiments. At the end of the binding experiments (24–96 h), the fibers were removed from the tubes and transferred to a defined volume of MeOH for extraction (volume varied between experiments). The fibers were extracted for 2–12 h. Overnight extraction was performed for lower volumes of MeOH. For HPLC measurements, the extract was diluted 1:1 with DI and internal standards were added. PBS solutions were diluted 1:1 with MeOH and internal standards were added. All solutions were vortexed for ≥ 15 s before HPLC measurements.

In experiments containing HSA, BSA, γ -globulins, or human serum, distribution coefficients were determined from the difference in the chemical amount sampled by the C18 fibers in the absence and presence of proteins (eq 1),³² provided that no other loss processes were observed in the recovery controls ($\geq 75\%$ recovery). The general equation used to calculate the protein–water distribution coefficient ($D_{\text{protein/w}}$, L kg^{-1}) was as follows:

$$D_{\text{protein/w}} = \frac{\frac{n_{\text{total}}}{n_{\text{C18}}} D_{\text{C18/w}} V_{\text{C18}} - V_{\text{w}} - D_{\text{C18/w}} V_{\text{C18}}}{m_{\text{protein}}} \quad (1)$$

where n_{total} (mol) is the total amount of PFAS added to the system, n_{C18} (mol) is the amount of PFAS sampled in the C18 fiber in the presence of the protein, $D_{\text{C18/w}}$ (L L^{-1}) is the C18 fiber–water distribution coefficient determined in the absence of protein, V_{w} (L) is the volume of water (i.e., PBS buffer) in the system, V_{C18} is the volume of the SPME fiber (520 nL), and m_{protein} is the mass of protein (kg) added to the test system. Binding of chemicals to proteins is typically reported as an association constant (K_{a} , M^{-1}), which is determined from the molar concentrations of bound and free chemicals and unbound proteins. Here, we report $D_{\text{protein/w}}$ because it is more convenient for mass balance calculations that involve phase volumes (eq 2). If binding of the chemical to the protein is well-below saturation, $D_{\text{protein/w}}$ can be converted to K_{a} using the molecular weight of the protein, as explained in detail in Endo and Goss.¹⁹

2.4. C18 Fiber Binding Experiments. We first performed distribution experiments between the C18 fiber and PBS buffer (i.e., protein-free water) using three perfluoroalkyl carboxylates (PFCA: perfluoroheptanoate: PFHpA, perfluorooctanoate: PFOA, and perfluorononanoate: PFNA), and two perfluoroalkyl sulfonates (PFSA: perfluorohexanesulfonate: PFHxS and perfluorooctanesulfonate: PFOS) as the test chemicals. To assess the reproducibility of distribution experiments, the C18–water distribution coefficient ($D_{\text{C18/w}}$) of PFOA was measured after 24 h of fiber–water incubation at 37 °C and 100 rpm horizontal agitation ($n = 10$). All subsequent experiments were performed at 37 °C and 100 rpm. $D_{\text{C18/w}}$ for PFHpA, PFOA, PFNA, PFHxS, and PFOS were determined after 0.5, 1, 2, 4, 8, 24, and 48 h of incubation ($n = 3$ per time point) to determine the time until fiber–water equilibrium was achieved. We tested for variability in $D_{\text{C18/w}}$ across three log units in water concentrations ($0.2\text{--}42 \mu\text{g L}^{-1}$) of the five PFAS to assess if saturation of the C18 fiber material occurred ($n = 3$ per concentration, 48 h incubation). $D_{\text{C18/w}}$ were determined for each PFAS in the 32 PFAS mixture after 48 and 96 h to confirm that fiber–water equilibrium was achieved for all study

chemicals ($n = 4$ per time point). Recoveries were measured in all experiments in vials containing PBS buffer only ($n = 3$ to 5) to confirm that no significant binding or loss processes other than to C18 fibers and dissolved proteins occurred.

2.5. Protein Binding Experiments. We compared distribution coefficients for PFHpA, PFOA, PFNA, PFHxS, and PFOS to bovine serum albumin ($D_{\text{BSA/w}}$) after 48 h incubations ($n = 4$) against previously measured literature values^{11,16} to validate the methodology used in this study. Fiber–water–protein equilibrium was assessed for the 32 PFAS by measuring $D_{\text{HSA/w}}$ after 48 and 96 h ($n = 4$ per time point). Notably, the distribution coefficients reported in this study were determined at molar ratios of PFAS to proteins ≤ 0.01 , which is approximately 100 times lower than the molar ratio at which saturation was observed for BSA.¹⁶ This should be considered when applying these distribution coefficients to systems with higher PFAS-to-protein ratios, such as *in vitro* toxicity tests. Binding to proteins was not expected to be rate-limiting based on previous work.³⁹ However, binding properties of proteins may change as denaturation of proteins, such as albumin and globulins occurs. Hence, our reported $D_{\text{protein/w}}$ were not measured in test durations of >48 h. Following method validation with BSA, distribution coefficients of the PFAS mixture to HSA ($D_{\text{HSA/w}}$), γ -globulins ($D_{\text{glob/w}}$), and human serum ($D_{\text{serum/w}}$) were measured ($n = 4$ per type of protein).

2.6. Chemical Analysis using HPLC-MS/MS. Targeted PFAS in this study were measured using an online SPE method on an Agilent (Santa Clara, CA) 6460 triple quadrupole liquid chromatography–tandem mass spectrometer at Harvard University.^{40,41} Additional information on chemical analysis is summarized in Section S4, SI. Linear and branched isomers were individually measured, and the reported distribution coefficients integrate the sum of isomers in the mixture.

2.7. Prediction of Binding to Human Serum. Empirically measured $D_{\text{HSA/w}}$ and $D_{\text{glob/w}}$ values for each PFAS were used to predict $D_{\text{serum/w}}$ (L L^{-1} , eq 2), f_{unbound} (eq 3), and C_{unbound} (mol L^{-1} , eq 4) in human serum by considering concentrations of globulins and serum albumin levels.

$$D_{\text{serum/w}} = D_{\text{HSA/w}} \text{VF}_{\text{HSA}} + D_{\text{glob/w}} \text{VF}_{\text{glob}} + \text{VF}_{\text{w}} \quad (2)$$

$$f_{\text{unbound}} = \frac{1}{\left(1 + D_{\text{HSA/w}} \frac{\text{VF}_{\text{HSA}}}{\text{VF}_{\text{w}}} + D_{\text{glob/w}} \frac{\text{VF}_{\text{glob}}}{\text{VF}_{\text{w}}}\right)} \quad (3)$$

$$C_{\text{unbound}} = f_{\text{unbound}} C_{\text{serum}} \frac{1}{\text{VF}_{\text{w}}} \quad (4)$$

where VF_{HSA} , VF_{glob} , and VF_{w} are the volume fractions of HSA, total globulins, and water in human serum (L L^{-1}), respectively. C_{serum} is the total chemical concentration in serum (mol L^{-1}). $D_{\text{HSA/w}}$ and $D_{\text{glob/w}}$ were transformed to volumetric units assuming protein densities of 1.36 and 1.20 kg/L for albumin and globulins, respectively. $D_{\text{w/w}}$ is assumed to be 1 (i.e., no differences in solubility are expected between the aqueous phase in human serum and pure water).

2.8. NHANES Survey Data. The U.S. Center for Disease Control and Prevention (CDC) conducts the NHANES survey every two years as a statically representative cross-sectional survey of the general population.⁹ We downloaded data on serum concentrations of PFAS, albumin, and total proteins from participants in the 2017–2018 NHANES survey cycle.

We used these data to quantify variability in blood protein concentrations among individuals, and how this affects f_{unbound} and C_{unbound} for different PFAS.⁹ The limit of detection (LOD) was 0.10 ng mL⁻¹ for serum PFAS, and 0.3 g dL⁻¹ for albumin and total proteins. Globulin concentrations were estimated from the difference between albumin and total protein concentrations. For measurements below detection, NHANES assigns a default value of the LOD $\times 1/\sqrt{2}$, which we retained. Blood proteins were measured as part of the standard biochemistry profile, while serum PFAS were measured in a one-third subsample of persons 12 years and over ($n = 1919$ individuals).

3. RESULTS AND DISCUSSION

3.1. Method Development and Validation. We observed good reproducibility (relative standard deviation, RSD = 1–11%) across replicate fiber–water distribution experiments (Table S2; Figures S3 and S4). Recoveries of 90–110% were observed for PFAS with $\eta_{\text{pfc}} \leq 9$ and 75–110% for PFAS with $\eta_{\text{pfc}} > 9$ (Figures S5 and S6). These results confirm negligible to small losses of PFAS due to binding or denaturation processes during fiber–water distribution experiments. Kinetic experiments with five PFAS showed equilibrium in the C18 SPME fibers was achieved within 48 h (Figure 1A). Rapid uptake of PFAS by the SPME fibers reduces the likelihood that biological materials degrade over the duration of the experiment.

$D_{\text{C18/w}}$ deviated by <30% between 48 and 96 h for all PFAS (Table S3). $D_{\text{HSA/w}}$ decreased between 48 and 96 h, particularly for $\eta_{\text{pfc}} > 9$ PFAS (Table S4). This may reflect denaturation of the proteins and hence reduced sorption of

PFAS to the specific binding sites of albumin.¹³ We therefore recommend performing protein and/or tissue binding experiments with the fibers within 48 h because binding to dissolved proteins is not expected to be rate-limited.⁴² We performed all subsequent experiments with a test duration of 48 h.

Linearity of sorption to fibers was observed for the five initial PFAS tested across a large range of concentrations (Figure 1B, slope (m) ≈ 1 for all tested chemicals). These results reinforce the utility of SPME C18 fibers for sampling PFAS across a wide range of concentrations expected in human blood and spiking experiments with mammalian tissues.

3.2. Binding to C18 Fibers, Serum Albumin, Globulins, and Human Serum. Binding to C18 fibers (i.e., log $D_{\text{C18/w}}$) increased with increasing molecular weight (M_w) of the PFAS up to ~ 600 g/mol (Figure 2A and Table S2). No differences were observed for PFAS with different functional groups (carboxylates, sulfonates, sulfonamides) or carbon chain structures (fluorotelomers) that had similar molecular weight. These results suggest that the sorption affinity of PFAS for the C18 polymer is predominantly size-dependent (i.e., dependent on the chain length of the perfluorinated tail and the size of the functional group of the molecule).

Bovine serum albumin (BSA) has been previously used as a surrogate for HSA in toxicokinetic models for PFAS.^{12,16,21} Our experiments reproduced literature values for $D_{\text{BSA/w}}$ within 0.5 log units (Figures S7 and S8). $D_{\text{HSA/w}}$ and $D_{\text{BSA/w}}$ agreed within 0.25 log units for 23 out of 27 PFAS. However, we found sorption to HSA was 0.5–1 log units higher than BSA for perfluorohexanoate (PFHxA), 5:3 fluorotelomer carboxylate (5:3 FTCA), perfluorobutanesulfonate (PFBS), and perfluoroheptanesulfonate (PFHpS) (Figure S9). This may reflect relatively more favorable electrostatic interactions with binding sites on HSA compared to those on BSA for PFAS with $\eta_{\text{pfc}} = 3$ –7. These results suggest $D_{\text{BSA/w}}$ may not be a good proxy for predicting the distribution to HSA for PFHxA, 5:3 FTCA, PFBS, and PFHpS.

PFAS binding to HSA ($D_{\text{HSA/w}}$) generally increased with higher M_w up until ~ 400 g/mol, followed by a plateau above this level (Figure 2B, Table S2). This result agrees with past mechanistic studies that showed a strong increase in HSA binding driven by electrostatic interactions (polar headgroup) and hydrophobic effects (nonpolar C–F chain) with PFAS carbon chain lengths up to $\eta_{\text{pfc}} = 8$.^{11,13} Both types of interactions are accommodated in HSA cavities for $\eta_{\text{pfc}} \leq 8$ PFAS. For $\eta_{\text{pfc}} > 8$ PFAS, reduced molecular flexibility limits their accessibility to the HSA binding pockets.¹³ A relatively higher $D_{\text{HSA/w}}$ for PFHpS compared to PFNA (Figure 2B) could be explained by the additional hydrogen bond established by the sulfonic headgroup.¹³

PFAS to γ -globulins differed from patterns observed for HSA. PFAS with a $M_w < 400$ exhibit relatively weak binding ($D_{\text{glob/w}} < 100$). Distribution coefficients for globulins (log $D_{\text{glob/w}}$) increased linearly for M_w between 400–600 g/mol but declined at $M_w > 600$ g/mol. γ -Globulins are larger than HSA (M_w of γ -globulins = 1193 kDa; M_w of HSA = 67 kDa) and are not known to bind with fatty acids. Binding of small molecules (<1 kDa) to blood globulins is generally considered to be of minor importance for the unbound fraction in human blood,⁴³ with the exception of $\alpha 1$ -acid glycoprotein binding of basic drugs like propranolol.⁴⁴ While γ -globulins possess binding sites for endogenous macromolecules like antigens ($M_w \approx 150$ kDa),⁴⁵ binding of <1 kDa molecules is thought to be driven by hydrophobic effects with the relatively large surface area of

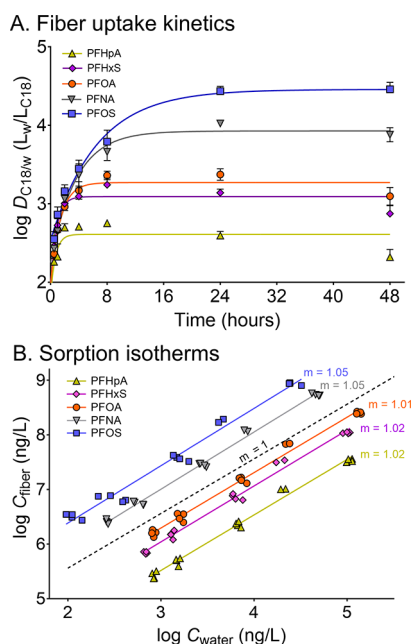


Figure 1. Results of solid-phase microextraction (SPME) C18 fiber-water distribution experiments for PFAS. Panel (A) shows concentration ratios in phosphate buffered saline (PBS) and fibers for five PFAS measured over 48 h in batch experiments ($n = 3$ per time point). The solid lines represent a first-order model fit. Panel (B) shows sorption isotherms for five PFAS and corresponding slopes of the linear isotherm fits ($n = 4$ per concentration), where m denotes the fitted slope of linear isotherms.

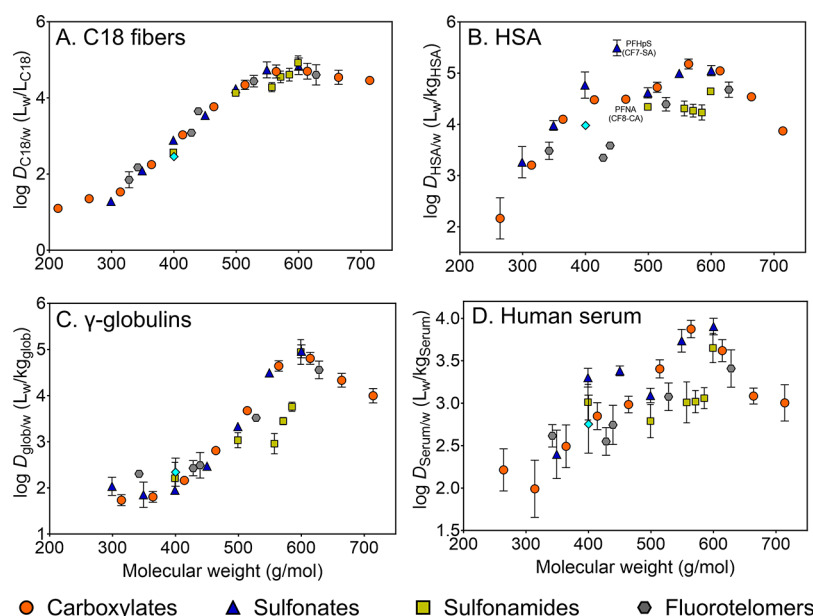


Figure 2. Variability in PFAS distribution coefficients to C18 fibers ($D_{C18/w}$), human serum albumin ($D_{HSA/w}$), γ -globulins ($D_{glob/w}$), and human serum ($D_{Serum/w}$) as a function of their molecular weight (M_w). Panels show distribution coefficients of PFAS to (A) the C18 fiber; (B) human serum albumin (HSA); (C) γ -globulins; and (D) human serum. PFAS classes are indicated by the color and shape of symbols with orange circles for carboxylates, blue triangles for sulfonates, yellow squares for sulfonamides, and gray hexagons for fluorotelomers. Turquoise diamonds = ADONA.

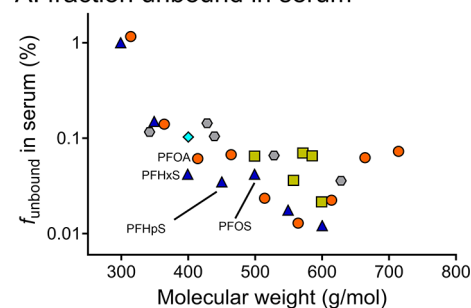
the protein.⁴⁶ This may explain the linear increase in $\log D_{glob/w}$ for PFAS of up to a M_w of 600 g/mol (Figure 2C), where a size-dependent cutoff occurs.

3.3. Distribution of PFAS in Human Serum. Binding to human serum measured using the C18 fiber method ($D_{Serum/w}$) increased linearly with M_w up to a peak of around 600 g/mol (Figure 2D). This trend is similar to that observed for HSA and γ -globulins (Figure 2B,C). Albumin appears to drive the distribution of PFAS with a $M_w < 500$ in the serum as illustrated by $D_{glob/w}/D_{HSA/w}$ ratios < 0.1 (Figure S10). The importance of PFAS binding to globulins (and related cotransport) increases with M_w until $D_{glob/w}/D_{HSA/w}$ ratios of ~ 1 for PFAS with $\eta_{pfc} = 12$ –13. Thus, globulins should be considered when assessing bioaccumulation of PFAS with $D_{glob/w}/D_{HSA/w}$ ratios of > 0.1 . HSA and globulins constitute the majority of serum biomolecules⁴⁷ but several other biomolecules, such as phospholipid-like fats, are present at lower concentrations and could also act as binding substrates for PFAS.^{48,49}

Values for $D_{Serum/w}$ suggest that PFAS are generally strongly bound to proteins in human serum ($f_{unbound}$ 0.01–1.2%). The estimated unbound fraction of PFAS decreased with an increasing molecular weight up to an inflection point at 600 g/mol, after which it increased again (Figure 3A). The unbound fraction of PFAS in serum was lower (0.035%) for PFHpS ($\eta_{pfc} = 7$ sulfonate), compared to PFOA ($\eta_{pfc} = 7$ carboxylate = 0.061%), PFHxS ($\eta_{pfc} = 6$ sulfonate = 0.041%), and PFOS ($\eta_{pfc} = 8$ sulfonate PFOS = 0.042%) (Figure 3A). We attribute this to the stronger relative binding of PFHpS to HSA (Figure 2B). Longer elimination half-lives in human serum have been reported for the PFAS with the greatest binding to serum proteins (lowest $f_{unbound}$) reported in this work (PFHpS $\approx$ PFHxS $>$ PFOS $>$ PFOA).^{26,50,51}

We tested whether distribution coefficients of PFAS to the most abundant blood proteins (albumin, globulins) reasonably predict the distribution of PFAS in human serum (i.e.,

A. fraction unbound in serum



B. experimental vs. predicted binding

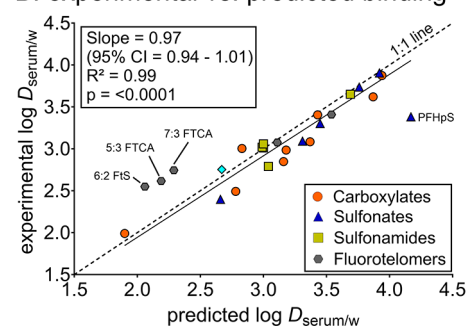


Figure 3. Panel (A) shows average fractions unbound of the study PFAS in the surrogate human serum (see SI section S1) measured with C18 fibers. Panel (B) shows the comparison of experimentally measured distribution coefficients of PFAS to serum ($\log D_{Serum/w}$) to distribution coefficients predicted by serum protein concentrations and distribution coefficients for globulins and human serum albumin (HSA). Serum protein concentrations were based on the average concentrations measured in the NHANES 2017–18 survey cycle ($n = 1919$). Turquoise diamond = ADONA.

$D_{Serum/w}$) (Figure 3B). Experimentally measured $D_{HSA/w}$ and $D_{glob/w}$ values were combined with average albumin and

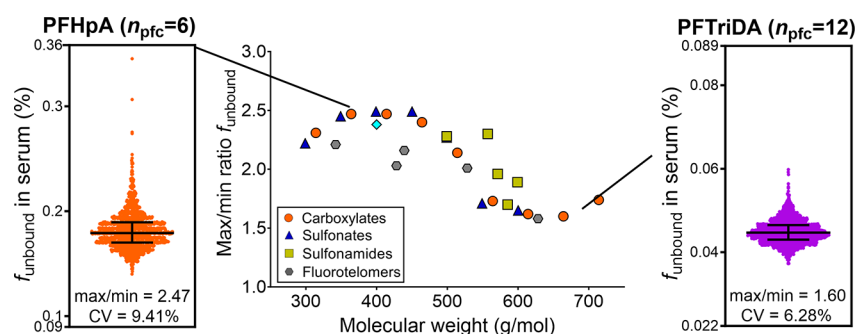


Figure 4. Variability in the unbound fraction of PFAS in serum (f_{unbound}). Center plot shows the range in f_{unbound} of PFAS for individuals participating in the NHANES 2017–18 survey cycle ($n = 1919$) predicted from measured albumin and globulin concentrations in blood and corresponding PFAS distribution coefficients. Left and right panels show the estimated f_{unbound} of perfluoroheptanoate (PFHpA) and perfluorotridecanoate (PFTrIDA) in serum for the same individuals. Black horizontal lines indicate the median and 75th and 25th percentiles of f_{unbound} . Coefficients of variation (CV) and maximum/minimum ratios of f_{unbound} are reported. Turquoise diamond = ADONA.

globulin blood levels reported in the NHANES 2017–18 survey cycle for this analysis ($VF_{\text{HSA}} = 4.1\%$, $VF_{\text{glob}} = 3.1\%$, $VF_w = 92.8\%$; Figure S11).⁹ Experimental log $D_{\text{serum/w}}$ agreed well with predicted values using eq 2 (Figure 3B, slope = 0.97, 95%-CI = 0.94–1.01, $R^2 = 0.99$). Three fluorotelomers (5:3 FTCA, 7:3 FTCA, 6:2 FtS) were underpredicted by the binding equation, suggesting that binding to other serum constituents (e.g., α -globulins, phospholipids) may be important. In contrast, PFHpS was overpredicted, potentially due to competitive binding between these PFAS and fatty acids contained in human serum. Good agreement between $D_{\text{serum/w}}$ that was directly measured and that predicted from protein binding for most PFAS reinforces the utility of blood protein concentrations for predicting their f_{unbound} and C_{unbound} in human populations.

3.4. Variability in PFAS Distribution among Individuals. We examined variability in the concentrations and fractions of unbound PFAS in blood across individuals based on differences in albumin and globulin concentrations measured in individuals participating in the 2017–18 NHANES survey cycle.⁹ Median NHANES serum concentrations were 41 g/L albumin and 31 g/L globulins, which is similar to serum protein levels reported for other populations.⁴⁷ Albumin and globulin concentrations ranged from 21 to 52 g/L and 20–55 g/L in the 2017–18 NHANES data, with coefficients of variation (CV) of 8.5 and 14%, respectively (Figure S12A). Individuals with lower globulin concentrations exhibited higher albumin concentrations (Figure S12B/C). The lowest observed albumin levels (below the 25th percentile or <29 g/L) may be related to common diseases like diabetes and fatty liver.^{29,52} Elevated serum globulins (Figure S12) have been associated with cardiovascular disease in other populations.³⁰

We estimated that differences in blood protein concentrations among individuals participating in the 2017–18 NHANES survey cycle contributed to a maximum variability in unbound PFAS fraction of 1.6 (for the 10:2 fluorotelomer sulfonate [10:2 FtS]) and 2.5 (PFHxS) (Figure 4).⁹ Compared to the first percentile, the 99th percentile fraction of unbound PFAS was a maximum of 1.3 times higher for perfluorotridecanoate (PFTrIDA) and 1.6 times higher for PFHxS. Coefficients of variation for f_{unbound} ranged from 6.3 (PFTrIDA) and 9.5% (PFHpS) (Figure S13). Variability in the unbound fraction of PFAS in serum was greatest for those with a M_w of <500 g/mol (factor of 2.0 to 2.5 variability; CV:

8.1–9.5%) but was smaller for those between $M_w \approx 500$ –600 g/mol (1.6 and 6.3%) (Figure 4, center plot).

An illustration of variability in the estimated unbound fractions for two PFAS in serum among NHANES individuals is shown in the left and right plots in Figure 4. Comparing PFHpA ($\eta_{\text{pfc}} = 6$; $M_w = 364$ g/mol) and PFTrIDA ($\eta_{\text{pfc}} = 12$; $M_w = 664$ g/mol), we see that relatively more individuals had a higher fraction unbound than the median for the shorter chain-length PFAS (PFHpA). The unbound fraction of PFHpA is mainly driven by serum albumin concentrations (i.e., $D_{\text{glob/w}}/D_{\text{HSA/w}}$ ratios are less than 0.1 for PFAS with $M_w < 500$ g/mol; Figure S11). In contrast, for PFTrIDA and other PFAS with higher molecular weights ($M_w > 500$ g/mol), we observed lower variability in the unbound fractions in serum (CV < 7%). This can be explained by competitive binding between albumin and globulins ($D_{\text{glob/w}}/D_{\text{HSA/w}}$ ratios 0.1–2; Figure S11). Albumin and globulin concentrations in serum exhibit an inverse pattern in the NHANES data set (Figure S13). PFAS binding to globulins is thus more important for individuals with albumin deficiency and vice versa. Variability in blood protein concentrations affect differences in the unbound fractions of shorter chain PFAS ($M_w < 500$ g/mol) more than for PFAS with higher molecular weights ($M_w > 500$ g/mol) (Figure 4). This could lead to greater interindividual variability in elimination half-lives and effects of exposures to PFAS with $M_w < 500$ g/mol. Substantial variability in elimination half-lives among individuals has been previously reported (factor of 5.6 for PFHxS and 3.5 for PFOS) and our results suggest variability in f_{unbound} may help to explain such differences.⁵³

Accounting for differences in serum protein concentrations can be helpful for interpreting the individual PFAS exposures. We used f_{unbound} to predict C_{unbound} from total serum concentrations of PFOA, PFHxS, PFNA, and PFOS measured in the 2017–18 NHANES survey cycle (eq 4).⁹ Lower albumin/globulin ratios (<1.4) of individuals could lead to higher bioavailable C_{unbound} of PFOA, PFHxS, PFNA, and PFOS relative to their total C_{serum} (up to 12% difference in ranks among individuals, Figures S14 and S15). Lower serum protein concentrations may enable more PFAS to reach molecular targets and trigger toxic effects.^{14–16} In contrast, individuals with higher protein concentrations and lower C_{unbound} may experience delayed PFAS elimination and prolonged exposures due to enhanced protein binding and recirculation of PFAS.^{17,18}

3.5. Implications for Toxicokinetic Modeling and Epidemiological Studies. Results of this study show that distribution coefficients for diverse PFAS to serum proteins and other blood constituents can be measured using the C18 SPME fiber method. Further, PFAS binding in human serum can be approximated by distribution coefficients ($D_{\text{glob/w}}$ and $D_{\text{HSA/w}}$) and concentrations of globulins and HSA. Less than 1.2% of PFAS were estimated to be unbound in human serum in this study because they are strongly bound to proteins. $M_w < 500$ PFAS mainly bind to albumin (fraction bound to albumin $> 98\%$). Binding to globulin becomes more significant as M_w increases, with approximately equal contributions of the two proteins to the total binding expected for $M_w > 600$ PFAS. Based on albumin and globulin concentrations of individuals, we estimated that f_{unbound} of PFAS varied by up to a factor of 2.5 among individuals participating in the 2017–18 NHANES survey cycle.⁹ While f_{unbound} of $M_w < 500$ PFAS is driven by the presence of albumin, lower variability in f_{unbound} of $M_w > 500$ PFAS is expected due to comparable protein binding strengths and inversely correlated concentrations of albumin and globulins in human serum.

The strong agreement between predicted and experimental $D_{\text{serum/w}}$ in this study suggests that our method is suitable for predicting the unbound fraction of PFAS in the serum of individuals. Given the simplicity and low costs of protein assays, serum proteins could be more routinely measured concurrently with PFAS exposures in human biomonitoring studies to develop such estimates at an individual level. While we compared predicted and experimental $D_{\text{serum/w}}$ for serum of a single donor, concurrent measurements of blood protein and PFAS concentrations in human biomonitoring studies could further evaluate the accuracy of our approach in predicting $D_{\text{serum/w}}$ and f_{unbound} . We used γ -globulins as a reference protein for the entire class of serum globulins. Future studies should aim to investigate variances in PFAS binding to other blood proteins to evaluate this assumption. Measuring additional serum proteins and biomarkers (e.g., hepatic enzymes and membrane transporters)⁵⁴ may explain the differences in predicted and experimental $D_{\text{serum/w}}$ observed for 6:2 FtS, 5:3 FtCA, 7:3 FtCA, and PFHpS (Figure 3B).

Our results suggest the relationship between external PFAS exposure and internal body burdens could be refined by accounting for variabilities in blood protein concentrations among individuals.^{55,56} Such refinements may also strengthen relationships between PFAS exposure and adverse health effects by improving estimates of internal dose and should be investigated in future work. Binding data reported in this study are useful for parametrizing physiologically based toxicokinetic (PBTK) models that can help elucidate the mechanisms underlying accumulation, elimination, and toxicity of PFAS. In combination with PBTK models, analyses of individual C_{unbound} could reveal the role of variable blood protein concentrations attributed to genetic, environmental, and longitudinal factors²⁸ to susceptibility to PFAS toxicity and elimination half-lives in humans.^{56,57} Diseases that alter blood protein levels, such as albumin deficiency in nonalcohol fatty liver disease,²⁹ could potentially trigger a feedback effect by allowing higher bioactive concentrations to reach already impaired target organs like a fatty liver and such hypotheses should be examined in future work. The SPME fiber method developed in this work could be used to investigate the binding of PFAS to tissue homogenates derived from target organs, such as the liver and kidneys. Resulting blood-tissue distribution coef-

ficients can be compared to organ concentrations in humans to assess the relevance of toxicokinetic processes beyond tissue binding such as enterohepatic recirculation and renal tubular reabsorption.^{58,59}

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.3c07415>.

Product specifications of the human serum measured in the study; photos of the experimental setup; list of study chemicals; summary of analytical methods; method validation and quality assurance; distribution coefficients measured in the study and comparison to literature values; temporal changes in distribution coefficients between 48 and 96 h incubation times; distribution of PFAS between different proteins in human serum; and distribution modeling based on the NHANES survey data (PDF)

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Notes

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Supporting Information

Binding of per- and polyfluoroalkyl substances (PFAS) to serum proteins:
Implications for toxicokinetics in humans

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S-1. Product specifications of the human serum measured in the study.

Product Name:

Human Serum – from human male AB plasma, USA origin, sterile-filtered

Product Number: H4522

MDL: MFCD00165829

Storage Temperature: -20 °C

TEST	Specification
Appearance (Color)	Colorless to Brown-Yellow to Brown
Appearance (Form)	Liquid
Cholesterol	110 - 210 mg/dl
Triglyceride	30 - 175 mg/dl
Glucose	60 - 140 mg/dl
Total Protein g/dl	4.0 - 9.0
pH	6.5 - 9.0
Sterility by USP guidelines	Pass
Endotoxin Level	≤ 10 EU/ml
Hemoglobin	≤ 30 mg/dl
Mycoplasma Test	None Detected
Osmolality	260 - 350
Expressed in MOSM/KG	
Tested For Infectious Agents	Tested
Sodium (Na)	100 - 200
Expressed in mmol/L	
Source	Conforms
Male plasma from donors within the United States of America	
Origin	Conforms
Processing country of origin: United States of America	
Iron (ug/dl)	35 - 180
Expiration Dated Product	-----

S-2. Photos of the experimental setup.

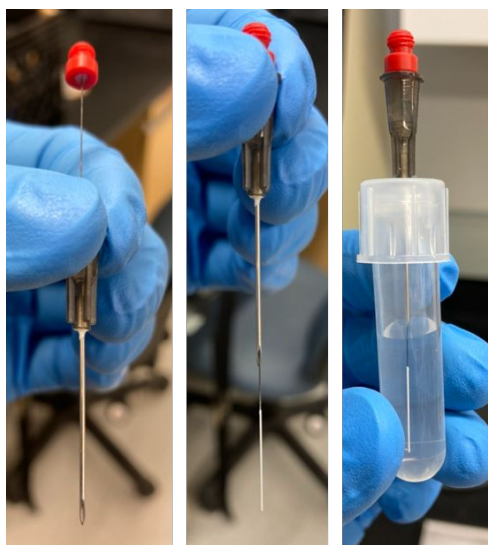


Figure S1. C18 SPME fibers used for binding experiments. The coating (white area) was completely immersed in PBS (+protein) and the syringes were stabilized by penetrating the vial cap.

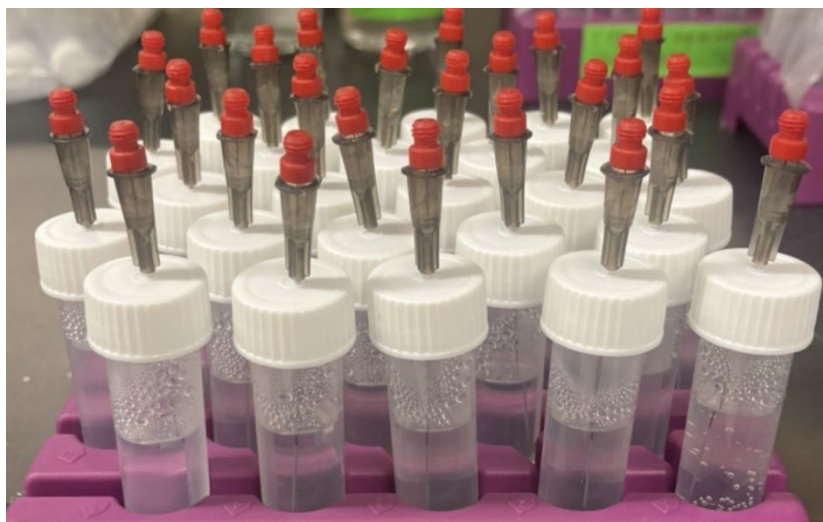


Figure S2. SPME fiber setup using polypropylene vials containing PBS and a defined amount of blood protein or human serum.

S-3. List of study chemicals.

Table S1. Chemicals tested in the 32 PFAS batch experiments and the respective isotopically labeled standards used for quantification.

Compound Name	Native Abbreviation	Isotope Labeled Abbrev.
Perfluoroalkyl Carboxylates		
Perfluorobutanoate	PFBA	MPFBA
Perfluoropentanoate	PFPeA	M5PFPeA
Perfluorohexanoate	PFHxA	M5PFHxA
Perfluoroheptanoate	PFHpA	M4PFHpA
Perfluorooctanoate	PFOA	M8PFOA
Perfluorononanoate	PFNA	M9PFNA
Perfluorodecanoate	PFDA	M6PFDA
Perfluoroundecanoate	PFUnDA	M7PFUnDA
Perfluorododecanoate	PFDoDA	MPFDoDA
Perfluorotridecanoate	PFTriDA	M2PFTeDA
Perfluorotetradecanoate	PFTeDA	M2PFTeDA
Perfluoroalkyl Sulfonates		
Perfluorobutane sulfonate	PFBS	M3PFBS
Perfluoropentane sulfonate	PFPeS	M3PFHxS
Perfluorohexane sulfonate	PFHxS*	M3PFHxS
Perfluoroheptane sulfonate	PFHpS	M8PFOS
Perfluorooctane sulfonate	PFOS*	M8PFOS
Perfluorononane sulfonate	PFNS	M8PFOS
Perfluorodecane sulfonate	PFDS	M8PFOS
Perfluoroalkyl Sulfonamides		
Perfluoro-1-octanesulfonamide	FOSA	M8FOSA
Perfluoro-1-hexanesulfonamide	FHxSA	M8FOSA
Perfluoro-1-butanesulfonamide	FBSA	M8FOSA
Perfluoro-1-decanesulfonamide	FDSA	M8FOSA
Perfluoroalkyl Sulfonamidoacetic Acids		
Perfluoro-1-octanesulfonamidoacetic acid	FOSAA	M8FOSA
N-methyl perfluoro-1-octanesulfonamidoacetic acid	N-MeFOSAA*	d3-N-MeFOSAA
N-ethyl perfluoro-1-octanesulfonamidoacetic acid	N-EtFOSAA*	d5-N-EtFOSAA
Fluorotelomer Sulfonates		
4:2 fluorotelomer sulfonate	4:2 FtS	M2-4:2 FtS
6:2 fluorotelomer sulfonate	6:2 FtS	M2-6:2 FtS
8:2 fluorotelomer sulfonate	8:2 FtS	M2-8:2 FtS
10:2 fluorotelomer sulfonate	10:2 FtS	M2-8:2 FtS
Fluorotelomer Carboxylates		
7:3 fluorotelomer carboxylate	7:3 FTCA	M5PFHxA
5:3 fluorotelomer carboxylate	5:3 FTCA	M5PFHxA
Polyfluoroalkyl ether carboxylates		
Dodecafluoro-3H-4,8-dioxananoate	ADONA	M8 FOSA

*Analyte exists in the standard as the linear and branched isomers.

S-4. Summary of analytical methods.

SPME fiber extracts and PBS samples were analyzed for 32 PFAS with an Agilent (Santa Clara, California, U.S.) 6460 triple quadrupole liquid chromatography-tandem mass spectrometer equipped with an Agilent 1290 Infinity Flex Cube online SPE located at Harvard University. PFAS concentrations were determined from a 12-point calibration curve ranging from 1 to 8,000 ng L⁻¹. Instrumental blanks (to monitor cross-contamination and carry-over) and the calibration curve were prepared with 50:50 MeOH:DI water with internal standard concentrations matching the samples. All analytes and internal standards were monitored for in electrospray ionization (ESI) negative mode. The linear and branched isomers of PFHxS and PFOS were quantified separately using calibration curves for each isomer. Initial conditions were 97% 2 mM ammonium acetate in water (A) and 3% 2 mM ammonium acetate in methanol (B). From 0.85 to 3.5 minutes the gradient was linearly increased to 54% B. From 3.5 to 16 minutes the gradient was linearly increased to 85% B and then linearly increased to 100% B until the end of the run (16.5 minutes).

S-5. Method validation and quality assurance.

Variability between replicates. Fiber-water distribution coefficients ($D_{\text{fiber/w}}$) of PFOA were measured based on the methods described in section 2.3. and 2.4. of the main manuscript. Fibers were incubated in PBS containing a PFOA for 24 hours. Fibers and PBS were extracted as described in the main manuscript. Resulting $D_{\text{fiber/w}}$ are presented in Figure S1.

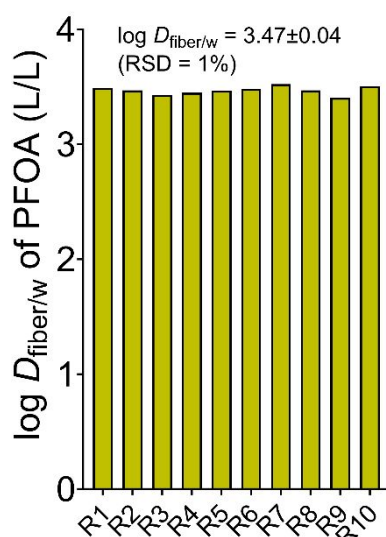


Figure S3. Fiber-water distribution coefficients ($D_{\text{fiber/w}}$) of PFOA measured in ten replicate vials were not significantly different, emphasizing the precision of the C18 fiber SPME method.

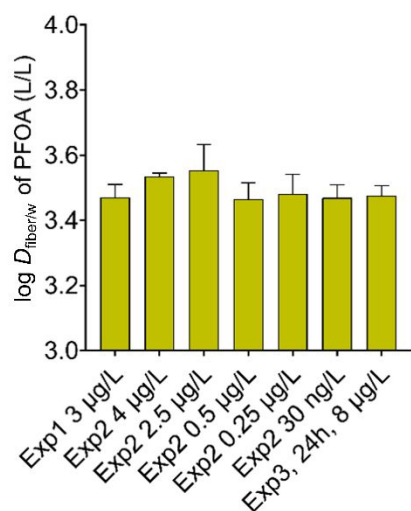


Figure S4. Fiber-water distribution coefficients ($D_{\text{fiber/w}}$) of PFOA measured in multiple distribution experiments at different chemical concentrations were not significantly different between experiments, emphasizing the reproducibility of the C18 fiber SPME method.

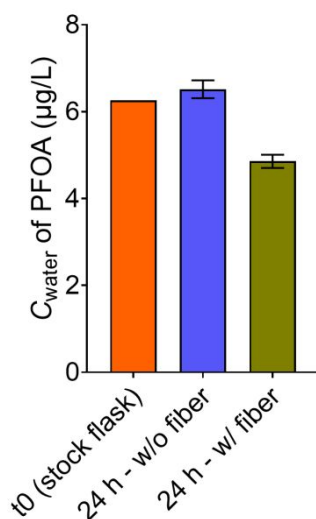


Figure S5. Chemical concentrations of PFOA in the water of the stock flask, in vials containing SPME fibers, and in control vials without SPME fibers after 24 hours of incubation. The results demonstrate that PFOA was not lost to other sorptive materials or degraded over the 24-hour experiment.

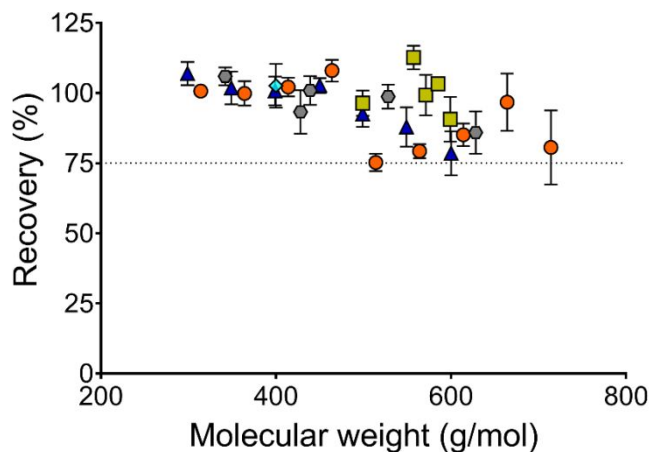


Figure S6. Recoveries of PFAS in PBS-fiber distribution experiments using the 32-PFAS batch. The dashed line represents 75% recovery, which was achieved for all chemicals tested. Turquoise diamond = ADONA.

S-6. Distribution coefficients measured in the study and comparison to literature values.

Table S2. Distribution coefficients of 32 PFAS to C18 fibers, human serum albumin (HSA), bovine serum albumin (BSA), γ -globulins (glob), and human serum, as measured in this study (n = 4). SE = standard error; N.D. = not detected because of too high or too low binding or insufficient method sensitivity.

Chemical	C-F chain length	Mol. weight (g/mol)	$\log D_{C18/w}$ ($\pm$ SE, L_w/L_{C18})	$\log D_{HSA/w}$ ($\pm$ SE, L_w/kg_{HSA})	$\log D_{BSA/w}$ ($\pm$ SE, L_w/kg_{BSA})	$\log D_{glob/w}$ ($\pm$ SE, L_w/kg_{glob})	$\log D_{serum/w}$ ($\pm$ SE, L_w/L_{serum})
Carboxylates							
PFBA	3	214.04	1.10 $\pm$ 0.05	N.D.	N.D.	N.D.	N.D.
PFPeA	4	264.05	1.35 $\pm$ 0.02	2.16 $\pm$ 0.40	N.D.	N.D.	2.21 $\pm$ 0.25
PFHxA	5	314.05	1.53 $\pm$ 0.06	3.20 $\pm$ 0.06	2.83 $\pm$ 0.14	1.73 $\pm$ 0.12	1.99 $\pm$ 0.34
PFHpA	6	364.06	2.25 $\pm$ 0.05	4.10 $\pm$ 0.05	3.93 $\pm$ 0.21	1.81 $\pm$ 0.12	2.49 $\pm$ 0.25
PFOA	7	414.07	3.03 $\pm$ 0.08	4.48 $\pm$ 0.04	4.20 $\pm$ 0.06	2.16 $\pm$ 0.08	2.85 $\pm$ 0.16
PFNA	8	464.08	3.77 $\pm$ 0.04	4.49 $\pm$ 0.08	4.54 $\pm$ 0.12	2.81 $\pm$ 0.05	2.99 $\pm$ 0.10
PFDA	9	514.08	4.34 $\pm$ 0.12	4.73 $\pm$ 0.10	4.81 $\pm$ 0.14	3.67 $\pm$ 0.05	3.40 $\pm$ 0.11
PFUnDA	10	564.09	4.69 $\pm$ 0.17	5.18 $\pm$ 0.10	5.17 $\pm$ 0.13	4.64 $\pm$ 0.11	3.87 $\pm$ 0.10
PFDoDA	11	614.10	4.70 $\pm$ 0.21	5.05 $\pm$ 0.07	5.12 $\pm$ 0.12	4.81 $\pm$ 0.13	3.62 $\pm$ 0.13
PFTriDA	12	664.10	4.54 $\pm$ 0.19	4.54 $\pm$ 0.05	4.47 $\pm$ 0.13	4.33 $\pm$ 0.15	3.08 $\pm$ 0.09
PFTeDA	13	714.11	4.46 $\pm$ 0.03	3.87 $\pm$ 0.07	3.87 $\pm$ 0.14	4.00 $\pm$ 0.16	3.00 $\pm$ 0.22
Sulfonates							
PFBS	4	299.09	1.28 $\pm$ 0.09	3.26 $\pm$ 0.30	2.28 $\pm$ 0.27	2.03 $\pm$ 0.20	N.D.
PFPeS	5	349.10	2.09 $\pm$ 0.08	3.98 $\pm$ 0.10	4.00 $\pm$ 0.26	1.85 $\pm$ 0.27	2.40 $\pm$ 0.29
PFHxS	6	399.11	2.89 $\pm$ 0.08	4.77 $\pm$ 0.26	4.61 $\pm$ 0.08	1.95 $\pm$ 0.08	3.30 $\pm$ 0.11
PFHpS	7	450.12	3.54 $\pm$ 0.05	5.49 $\pm$ 0.15	4.88 $\pm$ 0.19	2.47 $\pm$ 0.03	3.38 $\pm$ 0.06
PFOS	8	499.12	4.24 $\pm$ 0.11	4.62 $\pm$ 0.10	4.76 $\pm$ 0.10	3.33 $\pm$ 0.02	3.09 $\pm$ 0.08
PFNS	9	549.13	4.74 $\pm$ 0.21	4.99 $\pm$ 0.08	5.14 $\pm$ 0.12	4.49 $\pm$ 0.09	3.74 $\pm$ 0.13
PFDS	10	600.15	4.83 $\pm$ 0.22	5.05 $\pm$ 0.10	5.13 $\pm$ 0.09	4.96 $\pm$ 0.14	3.90 $\pm$ 0.10
Sulfonamides							
FBSA	4	299.11	N.D.	N.D.	N.D.	N.D.	N.D.
FHxSA	6	399.13	2.57 $\pm$ 0.04	N.D.	N.D.	2.21 $\pm$ 0.34	3.01 $\pm$ 0.21
FOSA	8	499.15	4.13 $\pm$ 0.08	4.34 $\pm$ 0.05	4.33 $\pm$ 0.07	3.03 $\pm$ 0.16	2.79 $\pm$ 0.20
FOSAA	8	557.18	4.28 $\pm$ 0.12	4.31 $\pm$ 0.14	4.30 $\pm$ 0.16	2.96 $\pm$ 0.22	3.01 $\pm$ 0.24
N-MeFOSAA	8	571.21	4.55 $\pm$ 0.15	4.27 $\pm$ 0.13	4.29 $\pm$ 0.13	3.45 $\pm$ 0.08	3.02 $\pm$ 0.13
N-EtFOSAA	8	585.24	4.61 $\pm$ 0.17	4.23 $\pm$ 0.15	4.29 $\pm$ 0.11	3.76 $\pm$ 0.10	3.06 $\pm$ 0.12
FDSA	10	599.16	4.93 $\pm$ 0.18	4.64 $\pm$ 0.06	4.69 $\pm$ 0.06	4.95 $\pm$ 0.27	3.65 $\pm$ 0.17
Fluorotelomers							
4:2 FtS	4	328.15	1.85 $\pm$ 0.21	N.D.	N.D.	N.D.	N.D.
5:3 FTCA	5	342.11	2.17 $\pm$ 0.09	3.48 $\pm$ 0.17	2.90 $\pm$ 0.27	2.30 $\pm$ 0.08	2.62 $\pm$ 0.13
6:2 FtS	6	428.17	3.08 $\pm$ 0.10	3.35 $\pm$ 0.05	3.44 $\pm$ 0.11	2.43 $\pm$ 0.17	2.55 $\pm$ 0.16
7:3 FTCA	7	439.10	3.65 $\pm$ 0.07	3.59 $\pm$ 0.07	3.79 $\pm$ 0.07	2.49 $\pm$ 0.28	2.74 $\pm$ 0.23
8:2 FtS	8	528.18	4.44 $\pm$ 0.15	4.39 $\pm$ 0.13	4.49 $\pm$ 0.15	3.52 $\pm$ 0.04	3.08 $\pm$ 0.16
10:2 FtS	10	628.20	4.60 $\pm$ 0.27	4.68 $\pm$ 0.15	4.88 $\pm$ 0.14	4.56 $\pm$ 0.19	3.41 $\pm$ 0.22
ADONA	6	400.05	2.46 $\pm$ 0.06	3.98 $\pm$ 0.04	3.84 $\pm$ 0.07	2.34 $\pm$ 0.30	2.75 $\pm$ 0.34

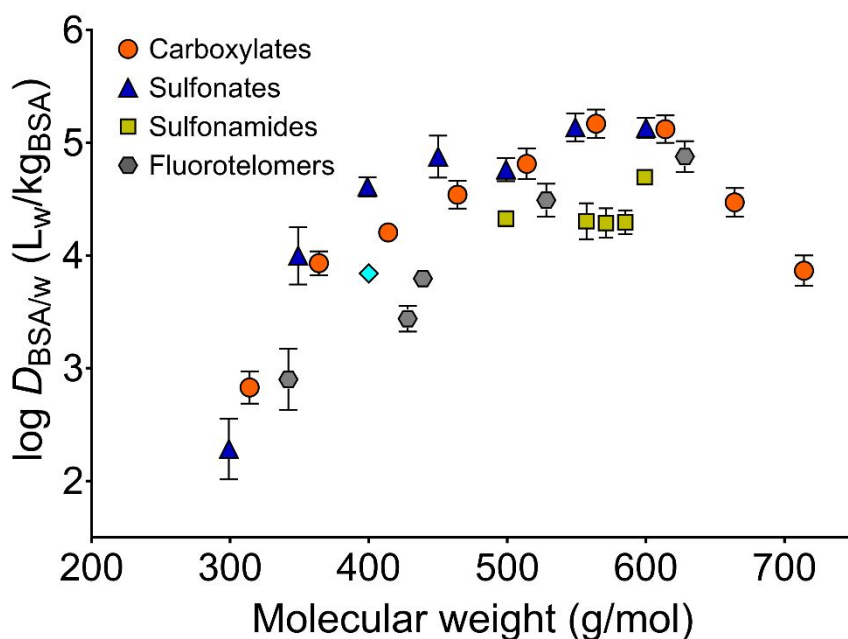


Figure S7. Distribution coefficients of the study PFAS to bovine serum albumin ($D_{BSA/w}$), plotted against the molecular weight of the chemicals. The chemicals are sorted according to their functional head-groups. Turquoise diamond = ADONA.

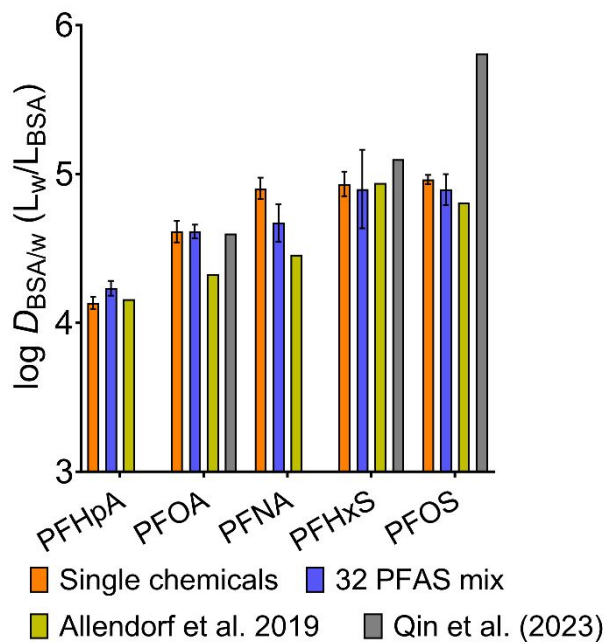


Figure S8. Bovine serum albumin-water distribution coefficients ($D_{BSA/w}$) of five PFAS measured as single chemicals in this study and in Allendorf et al. (2019) and Qin et al. (2023). $D_{BSA/w}$ measured in the 32 PFAS batch experiments are shown in comparison to emphasize consistency between the experiments and literature values.

S-7. Temporal changes in distribution coefficients between 48 and 96 hours incubation times.

Table S3. Distribution coefficients of 32 PFAS to C18 fibers measured after 48 and 96 hours of incubation (n = 4). SE = standard error; N.D. = not detected because of too high or low binding.

Chemical Abbr.	log $D_{C18/w}$ ($\pm$SE, L_w/L_{C18}) 48 hours	log $D_{C18/w}$ ($\pm$SE, L_w/L_{C18}) 96 hours	% deviation between time points (antilog)
PFBA	1.10 $\pm$ 0.05	1.14 $\pm$ 0.11	9.65%
PFPeA	1.35 $\pm$ 0.02	1.29 $\pm$ 0.15	-12.90%
PFHxA	1.53 $\pm$ 0.06	1.45 $\pm$ 0.21	-16.82%
PFHpA	2.25 $\pm$ 0.05	2.18 $\pm$ 0.24	-14.89%
PFOA	3.03 $\pm$ 0.08	2.95 $\pm$ 0.25	-22.38%
PFNA	3.77 $\pm$ 0.04	3.70 $\pm$ 0.23	-14.89%
PFDA	4.34 $\pm$ 0.12	4.38 $\pm$ 0.14	9.65%
PFUnDA	4.69 $\pm$ 0.17	4.77 $\pm$ 0.04	20.23%
PFDoDA	4.70 $\pm$ 0.21	4.65 $\pm$ 0.33	-10.87%
PFTriDA	4.54 $\pm$ 0.19	4.49 $\pm$ 0.24	-10.87%
PFTeDA	4.46 $\pm$ 0.03	4.35 $\pm$ 0.07	-22.38%
PFBS	1.28 $\pm$ 0.09	1.19 $\pm$ 0.30	-18.72%
PFPeS	2.09 $\pm$ 0.08	1.93 $\pm$ 0.17	-30.82%
PFHxS	2.89 $\pm$ 0.08	2.74 $\pm$ 0.13	-29.21%
PFHpS	3.54 $\pm$ 0.05	3.40 $\pm$ 0.20	-27.56%
PFOS	4.24 $\pm$ 0.11	4.17 $\pm$ 0.22	-14.89%
PFNS	4.74 $\pm$ 0.21	4.83 $\pm$ 0.13	23.03%
PFDS	4.83 $\pm$ 0.22	4.88 $\pm$ 0.11	12.20%
FBSA	N.D.	N.D.	N.D.
FHxSA	2.57 $\pm$ 0.04	2.53 $\pm$ 0.18	-8.80%
FOSA	4.13 $\pm$ 0.08	4.12 $\pm$ 0.10	-2.28%
FOSAA	4.28 $\pm$ 0.12	4.35 $\pm$ 0.23	17.49%
N-MeFOSAA	4.55 $\pm$ 0.15	4.67 $\pm$ 0.20	31.83%
N-EtFOSAA	4.61 $\pm$ 0.17	4.66 $\pm$ 0.20	12.20%
FDSA	4.93 $\pm$ 0.18	4.94 $\pm$ 0.22	2.33%
4:2 FtS	1.85 $\pm$ 0.21	1.81 $\pm$ 0.15	-8.80%
5:3 FTCA	2.17 $\pm$ 0.09	2.16 $\pm$ 0.22	-2.28%
6:2 FtS	3.08 $\pm$ 0.10	2.95 $\pm$ 0.25	-25.87%
7:3 FTCA	3.65 $\pm$ 0.07	3.65 $\pm$ 0.11	0.00%
8:2 FtS	4.44 $\pm$ 0.15	4.49 $\pm$ 0.06	12.20%
10:2 FtS	4.60 $\pm$ 0.27	4.61 $\pm$ 0.11	2.33%
ADONA	2.46 $\pm$ 0.06	2.31 $\pm$ 0.24	-29.21%

Table S4. Distribution coefficients of 32 PFAS to human serum albumin measured after 48 and 96 hours of incubation (n = 4). SE = standard error; N.D. = not detected because of too high or low binding.

Chemical Abbr.	log $D_{\text{HSA/w}}$ ($\pm$SE, $L_w/\text{kg}_{\text{HSA}}$) 48 hours	log $D_{\text{HSA/w}}$ ($\pm$SE, $L_w/\text{kg}_{\text{HSA}}$) 96 hours	% deviation between time points (antilog)
PFBA	N.D.	N.D.	N.D.
PFPeA	2.16 $\pm$ 0.40	2.13 $\pm$ 0.22	-6.67%
PFHxA	3.20 $\pm$ 0.06	3.23 $\pm$ 0.13	7.15%
PFHpA	4.10 $\pm$ 0.05	4.00 $\pm$ 0.16	-20.57%
PFOA	4.48 $\pm$ 0.04	4.52 $\pm$ 0.11	9.65%
PFNA	4.49 $\pm$ 0.08	4.48 $\pm$ 0.02	-2.28%
PFDA	4.73 $\pm$ 0.10	4.61 $\pm$ 0.05	-24.14%
PFUnDA	5.18 $\pm$ 0.10	5.09 $\pm$ 0.06	-18.72%
PFDoDA	5.05 $\pm$ 0.07	4.81 $\pm$ 0.10	-42.46%
PFTriDA	4.54 $\pm$ 0.05	4.29 $\pm$ 0.19	-43.77%
PFTeDA	3.87 $\pm$ 0.07	3.66 $\pm$ 0.16	-38.34%
PFBS	3.26 $\pm$ 0.30	2.88 $\pm$ 0.30	-58.31%
PFPeS	3.98 $\pm$ 0.10	4.12 $\pm$ 0.37	38.04%
PFHxS	4.77 $\pm$ 0.26	4.79 $\pm$ 0.16	4.71%
PFHpS	5.49 $\pm$ 0.15	5.44 $\pm$ 0.11	-10.87%
PFOS	4.62 $\pm$ 0.10	4.60 $\pm$ 0.09	-4.50%
PFNS	4.99 $\pm$ 0.08	4.71 $\pm$ 0.16	-47.52%
PFDS	5.05 $\pm$ 0.10	4.71 $\pm$ 0.05	-54.29%
FBSA	N.D.	N.D.	N.D.
FHxSA	N.D.	N.D.	N.D.
FOSA	4.34 $\pm$ 0.05	4.28 $\pm$ 0.04	-12.90%
FOSAA	4.31 $\pm$ 0.14	4.14 $\pm$ 0.10	-32.39%
N-MeFOSAA	4.27 $\pm$ 0.13	4.33 $\pm$ 0.14	14.82%
N-EtFOSAA	4.23 $\pm$ 0.15	4.25 $\pm$ 0.20	4.71%
FDSA	4.64 $\pm$ 0.06	4.79 $\pm$ 0.08	41.25%
4:2 FtS	N.D.	N.D.	N.D.
5:3 FTCA	3.48 $\pm$ 0.17	3.45 $\pm$ 0.11	-6.67%
6:2 FtS	3.35 $\pm$ 0.05	3.33 $\pm$ 0.05	-4.50%
7:3 FTCA	3.59 $\pm$ 0.07	3.52 $\pm$ 0.10	-14.89%
8:2 FtS	4.39 $\pm$ 0.13	4.23 $\pm$ 0.07	-30.82%
10:2 FtS	4.68 $\pm$ 0.15	4.28 $\pm$ 0.08	-60.19%
ADONA	3.98 $\pm$ 0.04	3.87 $\pm$ 0.07	-22.38%

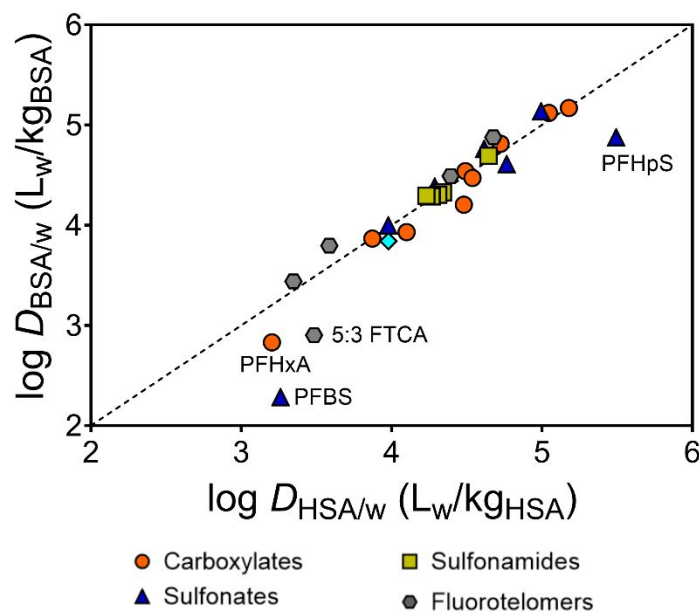


Figure S9. Bovine serum albumin- against human serum albumin-water distribution coefficients measured by the C18 fiber method. PFAS deviating from the 1:1 line are highlighted. Turquoise diamond = ADONA.

S-8. Distribution of PFAS between different proteins in human serum.

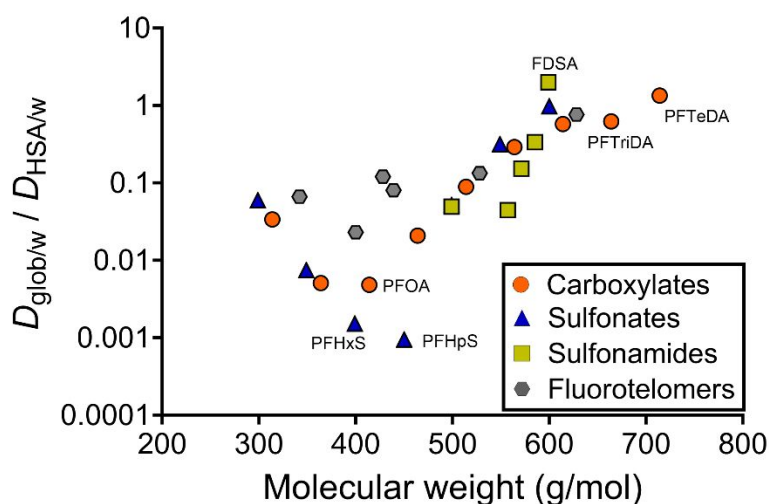


Figure S10. Ratios of globulin- and human serum albumin-water distribution coefficients of the study PFAS. The chemicals are sorted according to their functional head-groups. Individual PFAS with particularly high and low $D_{\text{glob/w}} / D_{\text{HSA/w}}$ ratios are highlighted. Turquoise diamond = ADONA.

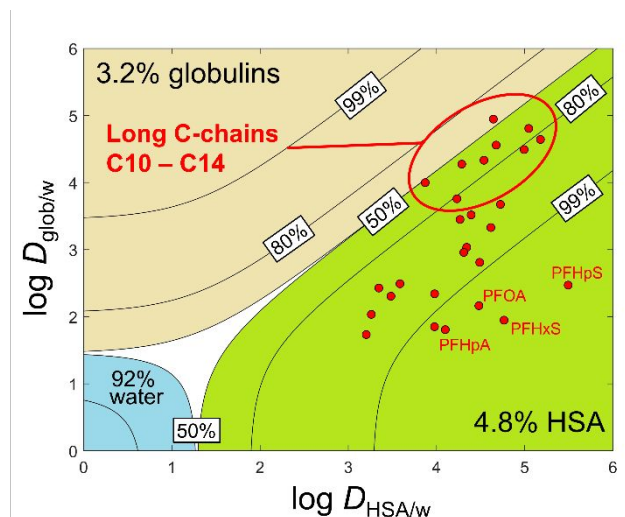


Figure S11. Chemical fractions bound to individual components of human serum, predicted assuming average volume fractions of 4.8% HSA, 3.2% globulins, and 92% water. Long C-chained PFAS are highlighted in the red circle and their shorter C-chained alternatives are highlighted as abbreviated chemical names.

S-9. Distribution modeling based on the NHANES survey data.

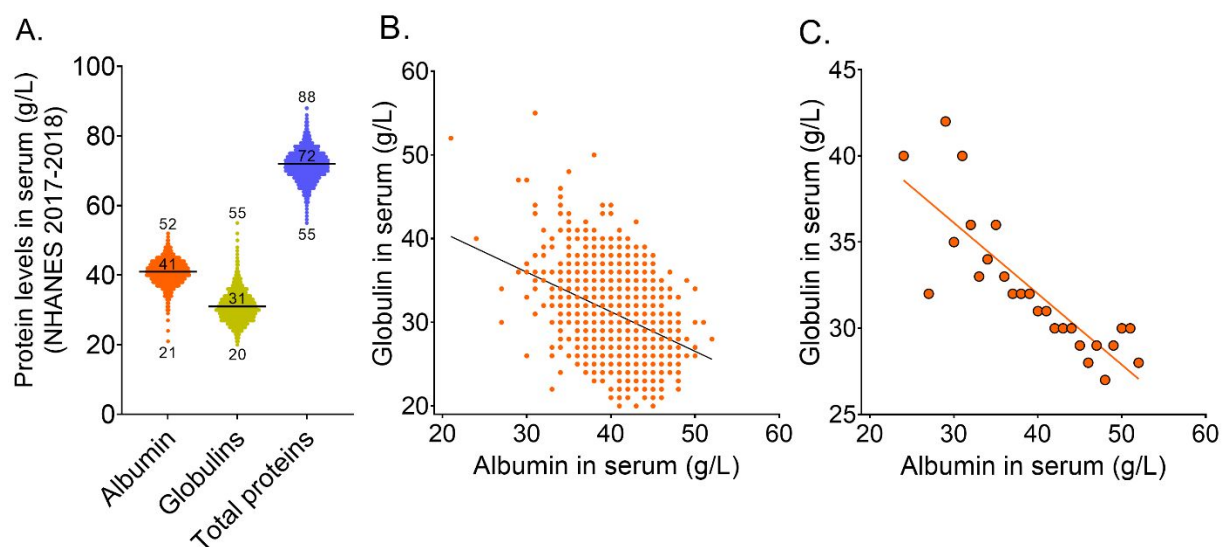


Figure S12. A. Albumin and globulin levels in human serum sampled in the years 2017-2018 of the NHANES survey (n = 1919). The black lines and numbers indicate the medians as well as minimum and maximum values. B. Albumin against globulin concentrations measured in the serum of 1919 individuals in the 2017-2018 NHANES survey cycle. The line shows the resulting linear regression. C. Average globulin concentrations for albumin bins (bin range = 21 – 55 g/L) measured in the serum of 1919 individuals in the 2017-2018 NHANES survey cycle. The solid line shows the resulting linear regression.

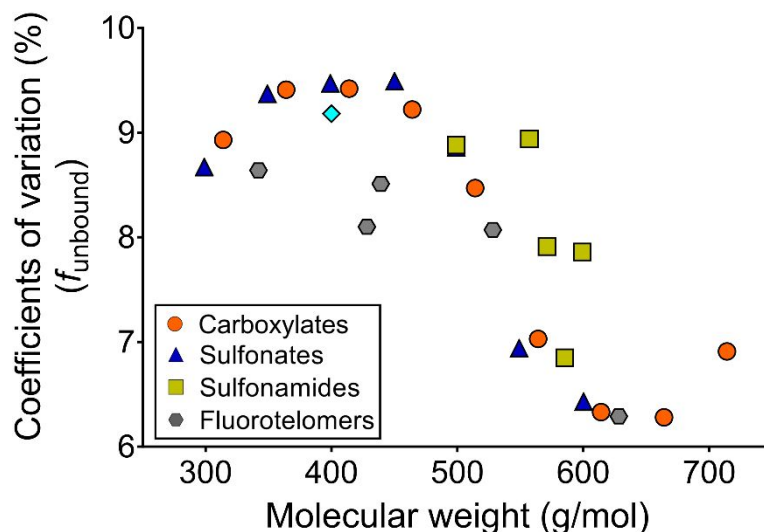


Figure S13. Coefficients of variation (%) of f_{unbound} of the study PFAS as predicted for individuals participating in the U.S. Center for Disease Control's National Health and Nutrition Examination Survey (NHANES 2017-2018 survey cycle; $n=1919$). Turquoise diamond = ADONA.

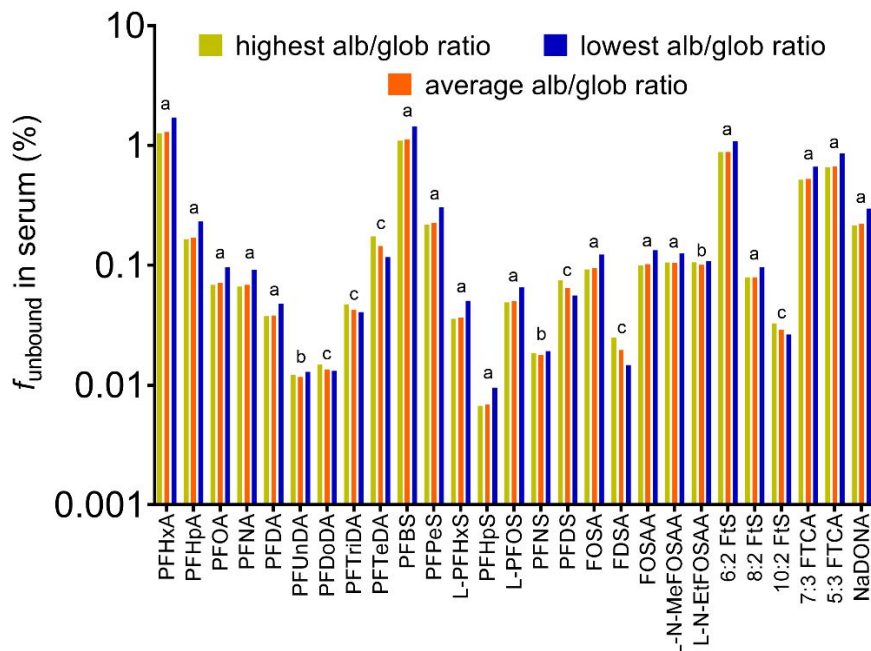
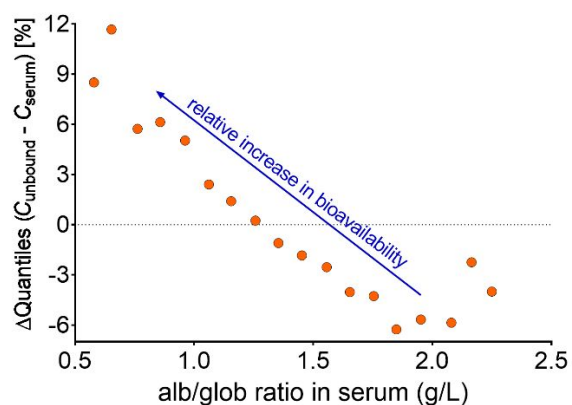
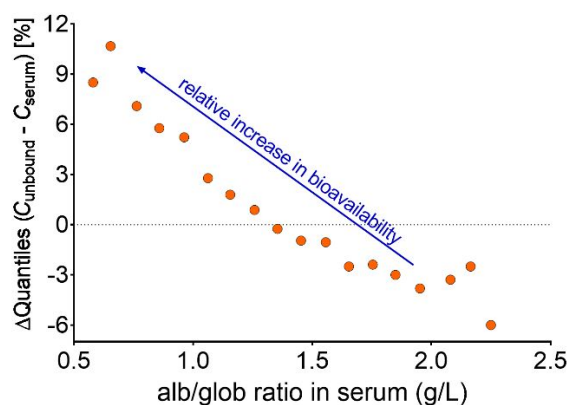


Figure S14. Minimum/maximum unbound fractions (f_{unbound}) of the study PFAS as predicted by individual albumin and globulin concentrations measured in the NHANES 2017-18 survey cycle ($n = 1919$) and corresponding distribution coefficients. a = funbound

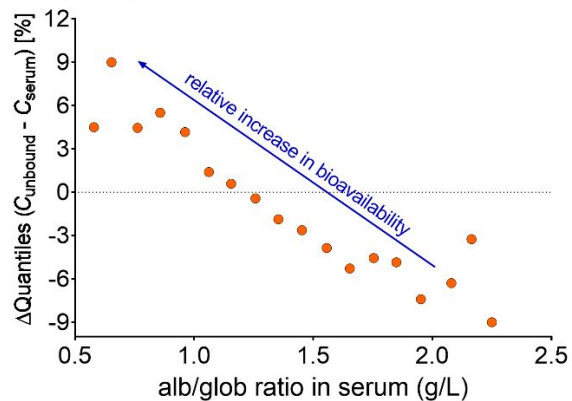
A. PFOA



B. PFHxS



C. PFNA



D. PFOS

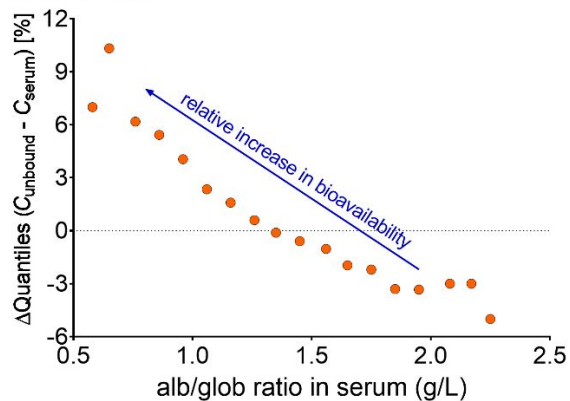


Figure S15. Differences in quantiles between C_{serum} and C_{unbound} of individuals as predicted by individual albumin and globulin concentrations measured in the NHANES 2017-18 survey cycle ($n = 1919$) and corresponding distribution coefficients.